

A piston cylinder contains 0.1 kg air at 100 kPa, 400 K which goes through a polytropic compression process with $n = 1.3$ to a pressure of 300 kPa. How much work has the air done in the process?

Process: $Pv^n = \text{Const.}$

Work integrated term:

$${}_1W_2 = \frac{1}{1-n} (P_2 V_2 - P_1 V_1)$$

